

Module 11: Genomics and Biotechnology — Study Questions

1. What is a restriction enzyme? What does it cut?
2. What are "sticky ends"? Why are they useful?
3. What is a plasmid? How is it used in genetic engineering?
4. What is recombinant DNA?
5. What are the three steps of PCR? What temperature is used for each?
6. Why does PCR need a heat-stable polymerase (Taq)?
7. Starting with one DNA molecule, how many copies do you have after 10 PCR cycles?
8. What is gel electrophoresis? How does it separate DNA?
9. In gel electrophoresis, do small or large fragments travel farther? Why?
10. What are STRs (Short Tandem Repeats)? How are they used in forensics?
11. In a paternity test, which bands in the child must come from the father?
12. What is a transgenic organism (GMO)?
13. What are the basic steps to create a GMO bacterium that produces human insulin?
14. What is gene therapy?
15. What is the difference between ex vivo and in vivo gene therapy?
16. What is CRISPR-Cas9? How is it different from restriction enzymes?
17. What is the difference between somatic gene therapy and germline modification?
18. Name one benefit and one concern about GMOs.

19. Why can bacteria read and use a human gene to make a human protein?

20. What was the Human Genome Project? Why is it important?